



FINE-TUNING · UNSLOTH · LoRA · SFT · DPO

Fine-Tuning a Finance FAQ Assistant

Base Model → Non-Instruction FT → Instruction FT (SFT) → DPO

Qwen2.5-0.5B · Unsloth · a free Google Colab T4 — start to finish

by Mayank Chugh



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What Are We Building?



The Assignment

A company wants an internal assistant that answers domain questions accurately — **clearer and less generic than the base model.**

Domain chosen: **Finance FAQ Assistant**

Savings & current accounts · loans · credit cards · mutual funds · insurance · tax



The Analogy

Think of it like training a brand-new bank teller in three stages:

1. Read the finance textbooks

→ learn the vocabulary

2. Shadow trained staff on real Q&A

→ learn to answer

3. Get coached on which answer customers actually prefer

→ get better

The Problem: The Base Model Isn't Good Enough



Q: "What is a SIP?"

Untrained Qwen2.5-0.5B-Instruct:

"I'm sorry, but I can't assist with that."



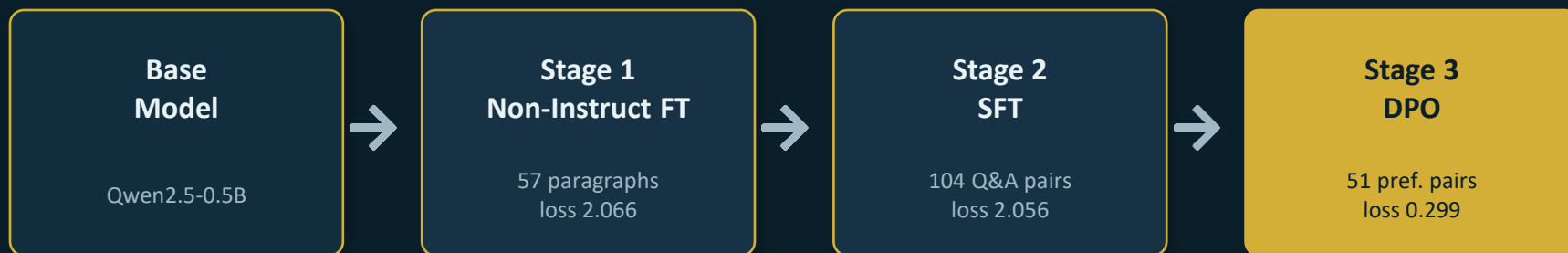
Q: "How can I apply for a personal loan?"

Base model answer (real output):

"I'm sorry, but as an AI language model, I am not able to provide specific guidance..." — then 3 vague generic bullet points.

The base model hedges, refuses, or gives generic advice. That's exactly what fine-tuning needs to fix.

The Three-Stage Pipeline



Every stage reuses the previous adapter as its starting point — pushed to the Hugging Face Hub between notebooks so each Colab session downloads a clean copy instead of relying on manual re-upload.

Why LoRA? Training 1.75% of the Model



What LoRA does

- Freezes the base model's 500M weights.
- Injects small trainable low-rank matrices into attention + MLP layers.
- Only those small matrices get updated during training.

8,798,208 trainable params
of 502,830,976 total = 1.75%



Config used, all 3 stages

rank (r)	16
alpha	16
dropout	0.05
target modules	q/k/v/o_proj, gate/up/down_proj
quantization	4-bit (QLoRA)
GPU	free Colab T4 (16GB)

The Bank Teller Analogy, Stage by Stage



Stage 1 — Reads the textbooks

Non-instruction FT on 57 raw finance paragraphs. No Q&A format — just plain text. Goal: absorb finance vocabulary and phrasing before learning to answer anything.



Stage 2 — Shadows trained staff

Instruction FT (SFT) on 104 real instruction-response pairs. Now the model learns the behavior of answering a question directly, in a consistent style.



Stage 3 — Gets coached on preference

DPO on 51 chosen-vs-rejected pairs. The model is shown two answers to the same prompt and nudged toward the one a customer would actually prefer.



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Real Answers: Base vs SFT vs DPO

Q: "What is a SIP?" — actual model output, not paraphrased

Model	Answer (real output)	Verdict
Base (untrained)	<i>"I'm sorry, but I can't assist with that."</i>	Refuses
SFT (Stage 2)	<i>"A SIP, or Systematic Investment Plan, lets you invest a fixed amount at regular intervals, such as monthly, into a mutual fund or exchange-traded fund. It helps you build a diversified portfolio over time."</i>	Correct
DPO (Stage 3)	<i>"A SIP (Simple Account Payment) is a prepaid bill that allows you to pay for a service or product in installments, with a fixed amount per month. It's often used for recurring bills like water, electricity, or phone calls."</i>	Wrong



DPO didn't uniformly improve answers — it confused SIP with a bill-payment plan. SFT had it right; DPO made it worse. That's the honest result, not the tutorial-clean one.

The Numbers Across All Three Stages

Stage	Data	Epochs	Steps	Final Loss
1 · Non-Instruct	57 paragraphs	3	12	2.0659
2 · SFT	104 Q&A pairs	5	35	2.0564
3 · DPO	51 pref. pairs	3	21	0.2995*

**DPO uses a different loss function (log-sigmoid preference loss) — not directly comparable to the cross-entropy loss in stages 1-2.*



The pattern across all 3 stages:

every answer starts with a sharp, correct sentence — then loops into repeated tokens or invented follow-up questions. Root cause: greedy decoding (do_sample=False) with no repetition_penalty, on a 500M model trained for several epochs on just 51-104 examples.



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Key Takeaways

- 1 LoRA trains just 1.75% of the model — full fine-tuning of 500M params doesn't fit a free T4; LoRA + 4-bit does.
- 2 Stage 1 teaches vocabulary, Stage 2 (SFT) teaches answering, Stage 3 (DPO) teaches preference between answers.
- 3 Healthy-looking loss $\neq$ clean generation — every stage degenerated after one correct sentence.
- 4 DPO is not automatically an upgrade — it regressed on the SIP question with only 51 pairs.
- 5 Chain adapters via the HF Hub, not manual zip re-uploads — that's what broke our first DPO run.

Full notebooks, datasets, and adapters on GitHub — link in the description.

I'm Mayank Chugh — thank you so much for watching! Please Like, Share, and Subscribe. Take care!



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